Codebook: r/soccer post corpus (2010–2026)

Version 3.0 (draft) — 23 October 2026 · r/soccer posts from the Arctic Shift archive, 1 January 2010 – 31 July 2026 · Assembled for the FIFA Communities chapter, McGill University.

This codebook documents every data file in `data/`, how each was produced, the meaning of every variable, and the known limitations of the corpus. It is written for co-authors, reviewers and any researcher who re-uses the data. Version 3.0 adds the **analysis variables** that the analytical pipeline appends to the master and yearly files (Section 4.9) and the methodological use of the data (Section 8), both specified in `docs/session-04_2026-10-22/01_analytical-pipeline-design.md`. **Status:** the data files on disk are still version 2.1 (42 columns) until stage S6 of that pipeline has run; this version is marked *draft* until then.

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1. Overview of the data files

The corpus is a **family of files derived from one raw export**. Use the *clean master* (or its yearly splits) for analysis; the *side files* preserve everything that was filtered out so that any decision can be audited or reversed.

File	Rows	Columns	Role
`data/soccer_posts.csv`	2,047,322	64	**Raw export.** Every post retrieved from Arctic Shift, de-duplicated by id. Never modified.
`data/soccer_data.clean.csv`	1,302,021	42	**Clean master.** Posts retained for analysis (63.6% of raw).
`data/yearly datasets/soccer.data.cleaned-YYYY.csv` (17 files, 2010–2026)	1,302,021 in total	42	The clean master split by calendar year (UTC). Identical format; convenient for memory-limited work.
`data/deleted_posts.csv`	283,433	43	Side file: posts deleted by their author (stage S2).
`data/removed_posts.csv`	375,613	43	Side file: posts removed by moderators, Reddit's spam filter, AutoModerator or takedowns (S3).
`data/bot_threads.csv`	48,156	43	Side file: template threads posted by bot accounts (S4).
`data/promotional_posts.csv`	34,974	43	Side file: promotional, gambling, merchandise, link-dump and stream-piracy posts (S5).
`data/encoding_issue_posts.csv`	335	43	Side file: posts whose text damage could not be repaired, or whose title is unintelligible (S6).
`data/non_english_posts.csv`	2,790	43	Side file: posts whose title is confidently non-English (S7).
`data/data artefacts/filter_summary.csv`	—	—	Count of rows per filter reason (audit table).
`data/data artefacts/spam_accounts.csv`, `piracy_accounts.csv`	—	—	Accounts caught by the account-level rules, with the statistics that triggered them.

File	Rows	Columns	Role
`data/data artefacts/encoding_repairs.csv`	—	—	Every title/body repaired in stage S6: post id, field, original text, repaired text, source encoding (126 fields).
`data/data artefacts/POSTS-*.jsonl`, `soccer_posts.csv.gz`, `jsonl_to_csv.py`	—	—	Collection artefacts: the three JSONL batches as downloaded, a compressed copy of the raw CSV, and the conversion script.
`data/analysis/post_variables.csv` *(planned, v3.0)*	1,302,021	11	**Analysis join table.** One row per clean post: the 8 analysis variables of Section 4.9 plus two audit columns (`topic_id_raw`, `category_id_agent`). Written by stage S5 of the analytical pipeline; the source of the columns appended to the master and yearly files in stage S6.
`results/analysis/topic_labels_all_bins.csv` *(planned, v3.0)*	one per topic	—	Two-agent label of every BERTopic topic (name, purpose, emotional tone, description, inclusion cues, debate rounds, agreement) with its `period_bin` and its discourse category.
`results/analysis/categories_all_bins.csv` *(planned, v3.0)*	one per category	—	The discourse taxonomy of each period bin: categories grouping similar topics, with the topics they contain.

Version 3.0 (planned). After stage S6 of the analytical pipeline, the clean master and the yearly files gain the 8 analysis columns of Section 4.9 (50 columns in total); the side files are unchanged (43 columns). The row counts do not change.

Accounting identity. raw = clean + deleted + removed + bot + promotional + encoding + non_english (2,047,322 = 1,302,021 + 283,433 + 375,613 + 48,156 + 34,974 + 335 + 2,790). Every post id appears in exactly one file. Verified by `code/03-data-validation.R` (28 checks).

Pipeline / provenance.

```
Arctic Shift API
  └─ fetch_posts_range.py ➔ POSTS-*.jsonl (3 batches)
       └─ jsonl_to_csv.py ➔ soccer_posts.csv (raw, 64 cols, de-duplicated by id)
            └─ 02-data-cleaning.R ➔ soccer_data.clean.csv + yearly files
                                    + 6 side files + audit tables
                 └─ 03-data-validation.R ➔ 28 checks (all pass for v2.1)
```

1.1 Clean rows per year

Year	Raw posts	Clean posts	Retained
2010	17,938	13,638	76%
2011	33,854	33,280	98%
2012	64,395	63,649	99%
2013	79,112	78,472	99%
2014	116,902	112,754	96%

Year	Raw posts	Clean posts	Retained
2015	126,399	107,461	85%
2016	137,766	85,318	62%
2017	160,413	105,531	66%
2018	197,865	126,743	64%
2019	185,591	118,664	64%
2020	168,357	61,794	37%
2021	173,373	65,286	38%
2022	155,575	64,584	42%
2023	130,421	70,334	54%
2024	116,843	76,473	65%
2025	108,138	74,348	69%
2026	74,380	43,692	59%

The retained share falls from 2020 onwards because Reddit's `removed_by_category` metadata only exists in the archive for posts from mid-2019 on; before that, moderator removals of link posts are not recorded and such posts cannot be identified. Analyses that compare periods should keep this in mind (Section 7).

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2. Data source and collection

- **Source.** Arctic Shift (https://arctic-shift.photon-reddit.com/), a community-run archive of Reddit that preserves the full post object returned by Reddit's API, including fields not shown in the Reddit interface.
- **Collection.** Three batches fetched on 9 August 2026 with `reddit-soccer-collector/src/fetch_posts_range.py`: 1 Jan 2010 – 31 Mar 2018, 31 Mar 2018 – 9 Jun 2026, 9 Jun 2026 – 31 Jul 2026 (the last batch overlaps the project's own live collector). Overlapping posts were de-duplicated by `id` during conversion to CSV; 2,047,322 unique posts remain.
- **Snapshot semantics.** Each record reflects the state of the post **when Arctic Shift's crawler retrieved it**, not when it was written. `score`, `num_comments`, flairs and removal status are therefore crawl-time values; a post crawled minutes after submission will show low engagement regardless of its later popularity. `retrieved_on` (raw file only) gives the crawl time.
- **Posts only.** The corpus contains submissions (titles, link URLs, self-text bodies). **No comment text is included**; `num_comments` is the only comment-related signal. 97.7 million comments are referenced by the raw posts.
- **Coverage.** Every month from 2010-01 to 2026-07 is present; monthly volumes are continuous across the batch boundaries. Dips reflect real events (e.g. the April–May 2020 COVID suspension).

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3. Cleaning and filtering rules

All cleaning is performed by `code/02-data-cleaning.R` in one pass over the raw file. Rules are applied **in the order below**; a post is assigned to the first rule it matches and written to that rule's side file with the exact `filter_reason` recorded. Counts are for version 2.1.

Stage	Side file	Rule (what is matched)	Rows
S2	`deleted_posts.csv`	`selftext == "[deleted]"` (`S2_selftext_deleted`)	234,217
S2	`deleted_posts.csv`	`removed_by_category ∈ {deleted, author}` (link posts deleted by the author) (`S2_removed_by_category_deleted`)	34,091
S2	`deleted_posts.csv`	title `[deleted by user]` (`S2_title_deleted_by_user`)	15,125
S3	`removed_posts.csv`	`selftext == "[removed]"` (`S3_selftext_removed`)	173,511

Stage	Side file	Rule (what is matched)	Rows
S3	`removed_posts.csv`	`removed_by_category == moderator` (`S3_removed_by_category_moderator`)	109,757
S3	`removed_posts.csv`	`removed_by_category == reddit` (Reddit spam filter) (`S3_removed_by_category_reddit`)	48,733
S3	`removed_posts.csv`	`removed_by_category == automod_filtered` (`S3_removed_by_category_automod_filtered`)	42,493
S3	`removed_posts.csv`	`removed_by_category == copyright_takedown` (`S3_removed_by_category_copyright_takedown`)	866
S3	`removed_posts.csv`	`removed_by_category == content_takedown` (`S3_removed_by_category_content_takedown`)	72
S3	`removed_posts.csv`	placeholder title (`[ Removed by moderator ]`, `[ Removed by Reddit ]`, …) (`S3_title_placeholder`)	1
S3	`removed_posts.csv`	body begins `[ Removed by Reddit on account of …` (`S3_selftext_removed_by_reddit`)	180
S4	`bot_threads.csv`	author ∈ {MatchThreadder, AutoModerator, 2soccer2bot, soccerbot} (`S4_bot_author`)	48,156
S5	`promotional_posts.csv`	account meets all three spam criteria (≥50 posts, ≥80 % one non-community domain, ≥90 % zero-comment) (`S5a_spam_account`)	16,396
S5	`promotional_posts.csv`	gambling/prediction domain tokens or betting-tip title phrases (`S5b_gambling`)	16,709
S5	`promotional_posts.csv`	cheap/replica jersey sellers (`four4two.com` etc.); questions exempt (`S5c_merch_spam`)	926
S5	`promotional_posts.csv`	title is only a URL (`S5d_url_only_title`)	364
S5	`promotional_posts.csv`	account whose every post is a "watch … live stream free/online/HD" link (`S5e_stream_piracy_account`)	579
S6	`encoding_issue_posts.csv`	U+FFFD replacement character (information lost upstream) (`S6_replacement_character`)	5
S6	`encoding_issue_posts.csv`	mojibake that no byte-level re-decoding repairs (`S6_mojibake_unrepairable`)	7
S6	`encoding_issue_posts.csv`	C0/C1 control characters that no byte-level re-decoding repairs (`S6_control_characters_unrepairable`)	209
S6	`encoding_issue_posts.csv`	title has no letter or digit (emoji/punctuation only) (`S6_title_without_letters_or_digits`)	114
S7	`non_english_posts.csv`	title confidently non-English (cld2 + cld3, evidence thresholds), body not confidently English (`S7_non_english`)	2,790
—	`soccer_data.clean.csv`	**retained**	**1,302,021**

3.1 Rule definitions and rationale

S1 Empty content. `title` and `selftext` both blank after trimming. Reddit requires a title, so no post matches; the rule is kept for completeness.

S2 Deleted (user-initiated). `selftext == "[deleted]"`, or `removed_by_category ∈ {deleted, author}`, or title `[deleted by user]`. The Pushshift/Arctic Shift convention is that `[deleted]` marks author deletion and `[removed]` marks moderator/platform removal (Baumgartner et al. 2020). For link posts the body is empty, so `removed_by_category` is the only reliable marker.

S3 Removed (moderator / platform). `selftext == "[removed]"`, or `removed_by_category ∈ {moderator, reddit, automod_filtered, copyright_takedown, content_takedown}`, or a placeholder title (`[ Removed by moderator ]`, `[ Removed by Reddit ]`, `[deleted]`, `[image processing failed]`), or a body beginning `[ Removed by Reddit`. Such posts were not part of the visible discourse at crawl time (`automod_filtered`

posts were never public; Reddit-spam-filtered posts have a median of 0 comments). Excluding removed content is standard practice in Reddit research (Proferes et al. 2021).

S4 Bot-authored threads. Author ∈ {`MatchThreadder`, `AutoModerator`, `2soccer2bot`, `soccerbot`}. These accounts post template threads (match threads with lineup/score blocks, daily discussion, free-talk and merchandise threads). The discourse they host lives in comments, which the corpus does not contain; their own text is boilerplate that would dominate a topic model. Human-posted match/post-match threads are **kept** and labelled `post_type = match_thread`.

S5 Promotional / non-discourse content.

- *S5a Spam accounts* — an account is classified as promotional when **all three** hold: (i) ≥ 50 posts in the raw corpus; (ii) ≥ 80 % of its posts link to a single domain outside the community-hosting set (reddit, twitter/x, youtube, imgur, streamable/streamja/streamin/streamff and other clip hosts, instagram, facebook, bsky, wikipedia); (iii) ≥ 90 % of its posts received zero comments. All posts by such accounts are moved. The three behavioural features — link-heavy posting, concentration on one URL/domain, and absence of interaction — are the feature families used in classic spammer-detection work on social platforms (Benevenuto et al. 2010; Stringhini et al. 2010; Lee et al. 2011). Those studies train classifiers; here the same signals are applied as transparent, auditable rules. The cut-offs are ours and were chosen from the empirical separation in this corpus: legitimate single-outlet accounts (e.g. newspaper accounts posting only their own domain) have 0–15 % zero-comment posts, content-farm accounts ≥ 95 %. 99 accounts match; they are listed in `data artefacts/spam_accounts.csv`.
- *S5b Gambling / betting-tip sites* — a non-community domain containing gambling tokens (prediksi, judi, taruhan, bandar, togel, casino, poker, sbobet, bet as a token or in bookmaker compounds, 303/88 gambling-brand patterns, Korean casino terms, …) or a title with betting-tip phrases. Mostly Indonesian "prediksi" (match-prediction) sites. A small number of genuine news items on bookmaker domains are caught (e.g. a poker-tournament result); they can be recovered from the side file by `filter_reason`.
- *S5c Merchandise spam* — "cheap/wholesale/replica … jersey/kit/shirt", `four4two.com`, `jerseymate`, "custom soccer uniforms". Titles containing a question mark are exempt so that genuine "where do I buy…?" questions are kept.
- *S5d URL-only titles* — the title consists solely of a URL (link dumps; mostly betting/affiliate links).
- *S5e Stream-piracy accounts* — accounts **all** of whose posts match the pattern "watch … live stream … free/online/HD" on a non-community domain (502 accounts, most with a single post; listed in `piracy_accounts.csv`). Isolated piracy posts by otherwise normal accounts are kept.

S6 Encoding-damaged or unintelligible text. Damage is detected when a title or body contains U+FFFD replacement characters, classic cp1252 mojibake (`Ã©`, `â€™`, `Â£`), or C0/C1 control characters (typically a garbled apostrophe such as `won\x92t`), or when the title has no letter or digit at all (emoji-only). Damaged fields are first **repaired where this can be done without guessing**: the text is converted back to the single-byte encoding it was wrongly decoded with (Windows-1252, ISO-8859-1 or Windows-1250, tried in that order) and the resulting bytes are decoded as UTF-8. A repair is accepted only if every character is representable in the candidate encoding, the bytes form strictly valid UTF-8 and no damage marker remains; the operation is a pure re-interpretation of the same bytes, is reversible, and is logged field by field in `data artefacts/encoding_repairs.csv` (126 fields in 126 posts, e.g. `Lucas Mouraâ€™s` → `Lucas Moura’s`, `Â£13.5million` → `£13.5million`). The validation script re-derives every original from its repaired value. Rows that cannot be repaired this way (for example an apostrophe followed by stray C1 bytes, or a lone U+FFFD) and emoji-only titles are moved to `encoding_issue_posts.csv` with the specific reason. The damage is inherited from Reddit/Arctic Shift, not introduced by the pipeline.

S7 Non-English titles. Language identification with Google's Compact Language Detector 2 (`cld2`) and CLD3 (`cld3`) on the **title**. A post is moved only when (a) cld2 is confident, (b) cld3 agrees, (c) the title has enough evidence — ≥ 15 letters with ≥ 8 outside the Latin script, or (for Latin-script languages) ≥ 30 letters, ≥ 5 word tokens and fewer than half of the tokens capitalised — and (d) the body, if present, is not confidently English. Condition (c) keeps goal-clip titles made of proper nouns ("Cagliari 1-[1] Lazio - Lucas Leiva"), which detectors mislabel as Italian/Dutch/Finnish; condition (d) keeps posts with an English title and a quoted foreign-language body. The decision is deliberately conservative: the clean master still contains

some non-English posts with short, name-heavy titles, and the `lang` column lets users apply their own threshold.

3.2 Content deliberately *not* filtered

- Posts by `[deleted]` authors whose text is intact (author account deleted later): 117,793 in v1 counts. Content integrity is preserved.
- Reposts of the same URL (33,416 URLs submitted 2+ times): normal community behaviour.
- Human-authored templated threads (pre/post/match threads): kept, labelled via `post_type`.
- Surveys, self-promotion of videos/blogs by otherwise engaged community members: keyword rules for these had unacceptable false-positive rates in testing and were not adopted.

3.3 Columns dropped from the raw export

The clean and side files keep 39 of the 64 raw columns and add three derived ones. The 25 dropped columns are either constant for a single-subreddit corpus, redundant, crawl-time artefacts or sparsely populated UI flags: `author_fullname`, `author_premium`, `author_is_blocked`, `upvote_ratio`, `created_utc` (duplicates `created_datetime`), `is_original_content`, `thumbnail`, `thumbnail_height`, `thumbnail_width`, `over_18`, `spoiler`, `stickied`, `locked`, `pinned`, `distinguished`, `contest_mode`, `no_follow`, `hide_score`, `archived`, `quarantine`, `subreddit`, `subreddit_id`, `subreddit_type`, `all_awardings`, `retrieved_on`. They remain available in `soccer_posts.csv` and can be joined back by `id`.

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4. Variable definitions (clean master, yearly and side files)

All 42 columns are stored as text; types below are the *intended* types after conversion. "Missing" means an empty cell. Examples are real values.

4.1 Identification

Variable	Type	Description	Example
`id`	string	Reddit post id (base-36). Unique across all files; join key to the raw file.	`88kf9q`
`name`	string	Reddit "fullname" (`t3_` + id).	`t3_88kf9q`
`permalink`	string	Path of the post on reddit.com.	`/r/soccer/comments/88kf9q/ray_wilkins_…/`

4.2 Content

Variable	Type	Description	Example
`title`	string	Post title, verbatim (UTF-8, may contain quotes and emoji). Never empty.	`Ray Wilkins: Ex-England captain in critical con`
`selftext`	string	Body of a self (text) post in Reddit Markdown; empty for link posts and, since 2022, optionally present on link posts. Non-empty for ~10.6% of clean posts. May contain line breaks.	`The votes are in, you decided…`

Variable	Type	Description	Example
`url`	string	Linked URL for link posts; the post's own Reddit URL for self posts.	`https://www.bbc.co.uk/sport/…`
`domain`	string	Domain of `url` (`self.soccer` for self posts).	`twitter.com`
`is_self`	boolean (`True`/`False`)	Self (text) post vs link post.	`False`
`is_video`	boolean	Post hosts a Reddit-native video (`v.redd.it`). Missing for older posts.	`False`
`is_reddit_media_domain`	boolean	Linked content hosted on Reddit's media CDN. Missing for older posts.	`True`
`post_hint`	string	Reddit's content-type hint: `link`, `image`, `hosted:video`, `rich:video`, `self`, `gallery`. Missing for ~60 % of posts (older posts, self posts).	`rich:video`
`media`	JSON string	Embedded-media metadata (`reddit_video`, oEmbed data). Empty when none.	`{"reddit_video":{…}}`
`preview`	JSON string	Preview image URLs and resolutions. Empty when none.	`{"images":[{…}]}`

4.3 Author

Variable	Type	Description	Example
`author`	string	Username. `[deleted]` when the account was deleted or the post removed after crawl.	`whistledick`
`author_flair_text`	string	Author's subreddit flair (usually a club; emoji-code form `:FC_Porto:` in later years).	`Arsenal`

Variable	Type	Description	Example
`author_flair_css_class`	string	CSS class of the author flair (encodes team/country ids in older years).	`s1 1 team-1 country-eng`
`author_flair_background_color`, `author_flair_text_color`, `author_flair_type`, `author_flair_template_id`	string	Styling and template metadata of the author flair; mostly missing before 2018.	`dark`, `richtext`
`author_flair_richtext`	JSON string	Rich-text components of the author flair (`[]` when none).	`[{"e": "text", "t": "Arsenal"}]`

4.4 Post flair

Variable	Type	Description	Example
`link_flair_text`	string	Post category assigned by moderators/author. Labels drift over time (e.g. `Official source`, `Official Source`, `:n_official: Official Source`); normalise before counting. Missing for ~half of posts.	`Media`
`link_flair_css_class`, `link_flair_background_color`, `link_flair_text_color`, `link_flair_type`, `link_flair_template_id`	string	Styling and template metadata of the post flair.	`#5b5b5b`
`link_flair_richtext`	JSON string	Rich-text components of the post flair (`[]` when none).	`[{"e":"emoji","u":"…"}]`

4.5 Engagement (crawl-time values)

Variable	Type	Description	Example
`score`	integer	Net votes at crawl time (Reddit "fuzzes" scores).	`10`
`ups`	integer	Equals `score` in modern Reddit data.	`10`
`downs`	integer	Always 0 or missing (Reddit stopped exposing downvotes). Uninformative.	`0`
`num_comments`	integer	Number of comments at crawl time. Only comment-related variable in the corpus.	`142`
`num_crossposts`	integer	Times the post was cross-posted *to other subreddits*. Missing for older posts.	`0`
`total_awards_received`, `gilded`	integer	Award counts (awards existed 2018–2023). Mostly 0/missing.	`0`

4.6 Time

Variable	Type	Description	Example
`created_datetime`	datetime (UTC)	Submission time, `YYYY-MM-DD HH:MM:SS`, UTC. Derived from the raw Unix timestamp `created_utc`.	`2018-03-31 18:02:01`
`year`	integer (derived)	Calendar year of `created_datetime` (UTC). Used to split the yearly files.	`2018`

Variable	Type	Description	Example
`edited`	mixed	`False` if never edited; a Unix timestamp (seconds) of the last edit; `True` for some pre-2012 posts where Reddit did not store the time.	`False` / `1522520000`

4.7 Moderation and subreddit

Variable	Type	Description	Example
`removed_by_category`	string	Removal category recorded by Reddit (`moderator`, `reddit`, `automod_filtered`, `deleted`, `author`, `copyright_takedown`, `content_takedown`). **Always empty in the clean master** (all such posts are in the side files); populated only from mid-2019.	`moderator`
`subreddit_subscribers`	integer	r/soccer subscriber count at crawl time. Missing before ~2018.	`903813`

4.8 Derived classification columns (added by the pipeline)

Variable	Type	Description	Values
`post_type`	string	Structural type of the post from title/flair patterns: `match_thread` (pre-/post-/match threads), `daily_thread` (daily discussion, free-talk, Monday Moan, etc.), `regular` (everything else). Informational only; nothing is filtered on it.	`regular`
`lang`	string	ISO-639-1 code of the language detected by cld2 on title + body; `und` when the detector was not confident. Informational; the S7 filter uses a stricter, title-based rule (Section 3.1).	`en`
`filter_reason`	string	**Side files only.** The rule that moved the row, e.g. `S3_removed_by_category_moderator`, `S5a_spam_account`, `S7_non_english`.	—

4.9 Analysis variables (appended by the analytical pipeline; version 3.0, planned)

These eight columns are produced by the analytical pipeline designed in session 04 (`docs/session-04_2026-10-22/01_analytical-pipeline-design.md`) and appended, after the 42 columns above, to the clean master and the yearly files in its final stage (S6). They are **model outputs**, not properties of the post: topics come from a BERTopic model fitted separately to each period bin, categories from two Qwen3-1.7B "analyst" agents that must agree on each label, and sentiment from the VADER lexicon. Side files do not carry them. Provenance of every value is recorded in `data/analysis/post_variables.csv` (plus two audit columns) and in the per-bin run folders `results/analysis/bin-N/`.

Variable	Type	Description	Values
`period_bin`	integer	Analysis period of the post, from the UTC calendar date of `created_datetime`. Bins: 1 = 2010-01-01 – 2010-09-30 (2010 World Cup window); 2 = 2010-10-01 – 2014-01-31; 3 = 2014-02-01 – 2014-09-30 (2014 WC); 4 = 2014-10-01 – 2018-01-31; 5 = 2018-02-01 – 2018-09-30 (2018 WC); 6 = 2018-10-01 – 2022-06-30; 7 = 2022-07-01 – 2023-02-28 (2022 WC); 8 = 2023-03-01 – 2026-01-31; 9 = 2026-02-01 – 2026-07-31 (2026 WC). Contiguous; every post has exactly one bin.	`1`–`9`
`period_type`	string	`tournament` for bins 1, 3, 5, 7, 9; `interim` for bins 2, 4, 6, 8. Derivable from `period_bin`; a convenience column.	`tournament` / `interim`
`topic_id`	integer	BERTopic topic of the post **within its period bin** (one model per bin; ids are bin-local, so the key is `period_bin` + `topic_id`). After thresholded outlier reassignment: `-1` = outlier (no topic); empty = not modelled (fewer than 3 word tokens in title + body).	`-1`, `0`, `1`, … (bin-local); empty

Variable	Type	Description	Values
`topic_assignment`	string	How `topic_id` was obtained: `cluster` = assigned by HDBSCAN; `reassigned` = outlier attached to its nearest topic because the embedding similarity reached the calibrated threshold; `outlier` = below the threshold, `topic_id = −1`; `not_modelled`.	`cluster` / `reassigned` / `outlier` / `not_modelled`
`category_id`	string	Discourse category of the post's topic in the two-agent taxonomy of its period bin (after co-author overrides); bin-local id (key: `period_bin` + `category_id`). Empty for outliers and not-modelled posts. Definitions in `results/analysis/categories_all_bins.csv`.	`cat_1`, `cat_2`, … (bin-local); empty
`post_sentiment`	float	VADER compound sentiment score of the cleaned **title** (the unit every post has; comparable across bins regardless of body length). 4 decimals.	−1.0000 … 1.0000
`post_sentiment_full`	float	VADER compound score of title + body (body truncated to 1,000 characters); a second measure for analyses of self posts. 4 decimals.	−1.0000 … 1.0000
`post_sentiment_label`	string	Categorised `post_sentiment` with VADER's conventional thresholds: `positive` > 0.05, `negative` < −0.05, otherwise `neutral`.	`positive` / `neutral` / `negative`

Audit columns in `post_variables.csv` only: `topic_id_raw` (the post's HDBSCAN cluster, re-numbered by cluster size — i.e. the BERTopic topic id before outlier reassignment; HDBSCAN's own labels and the mapping are kept in `results/analysis/bin-N/`) and `category_id_agent` (the category before co-author overrides). The per-topic labels (`results/analysis/topic_labels_all_bins.csv`) carry, for each `period_bin` × `topic_id`: `name`, `purpose`, `emotion_tone`, `tone_category` (controlled vocabulary), `description`, `inclusion_cues`, `round_iterations` (debate rounds until the two agents agreed), `agreed`, `agreement_mode` (`identical`, `accepted_by_A`, `accepted_by_B`, `stalemate`, `cap`, `no_answer`), `needs_review`, both agents' final answers, the category, sizes (`n_cluster`, `n_total`, `share`), keywords and the model/prompt versions.

---

5. File format conventions

- **Encoding / layout.** UTF-8, comma-separated, header row, RFC-4180 quoting (fields containing commas, quotes or line breaks are wrapped in double quotes; embedded double quotes are doubled). `selftext` frequently spans multiple lines — use a CSV parser, not line-based tools, to count rows.
- **Missing values** are empty cells. There are no `NA`/`NULL` strings.
- **Booleans** are the strings `True` / `False` (as delivered by Reddit).
- **Timestamps** are UTC.
- **JSON columns** (`author_flair_richtext`, `link_flair_richtext`, `media`, `preview`) hold serialised JSON; parse with `jsonlite::fromJSON()` (R) or `json.loads()` (Python). They are empty or `[]` when no data exists.
- **All columns are text.** Recommended readers:

```
# R
library(readr)
d <- read_csv("data/soccer_data.clean.csv",
              col_types = cols(.default = col_character()), na = character())
d$num_comments     <- as.integer(d$num_comments)
d$created_datetime <- as.POSIXct(d$created_datetime, tz = "UTC")
```

```
# Python
import pandas as pd
d = pd.read_csv("data/soccer_data.clean.csv", dtype=str, keep_default_na=False)
```

- **Warning.** Do not load these files with `data.table::fread()` default settings: on this corpus it fails to un-escape doubled quotes and silently corrupts every quoted field (this bug affected version 1 of the

cleaned files; see Section 9).
- All files (master, yearly, side files) are written by the same `readr::write_csv()` call and share exactly the same conventions.

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6. Dataset statistics (clean master, v2.1)

Statistic	Value
Posts	1,302,021
Date range (UTC)	2010-01-01 02:34:08 – 2026-07-31 23:32:20
Self (text) posts	12.6%
Posts with non-empty `selftext`	10.6%
Median title length (characters)	62
Posts with zero comments at crawl	15.2%
Posts by `[deleted]` authors (content intact)	113,928
`post_type`: match_thread / daily_thread / regular	46,958 / 1,780 / 1,253,283
`lang`: en / und / other	1,078,542 / 143,057 / 80,422

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7. Known limitations for analysis

1. **Posts only.** No comment text. Discourse that takes place in comment threads (the bulk of r/soccer interaction, including inside match threads) is not observable; `num_comments` gives only its volume.
2. **Title-dominated text.** About 88 % of clean posts are link posts whose only text is the title (median 62 characters). Topic models and sentiment measures mostly operate on short, headline-like text; results should be interpreted accordingly.
3. **Crawl-time snapshot.** Scores, comment counts and flairs reflect the moment of archiving. Early-crawled posts under-report engagement.
4. **Removal metadata is era-dependent.** `removed_by_category` exists only from mid-2019. Before that, moderator-removed *link* posts cannot be identified and remain in the clean master; from 2019 on they are removed. Year-to-year comparisons of volume or of removed shares must account for this.
5. **Deleted authors.** 117,793 clean posts carry `author = [deleted]`; author-level analyses lose these posts.
6. **Language.** The S7 filter is conservative (Section 3.1); short, name-heavy non-English titles remain. Use `lang` for stricter subsetting.
7. **Encoding.** 126 posts had mangled characters repaired by byte-level re-decoding (logged in `encoding_repairs.csv`); 335 posts that could not be repaired without guessing, or whose title is emoji-only, are in `encoding_issue_posts.csv`.
8. **Flair drift.** Flair labels and their spelling change over the 16 years; normalise before using them as categories.
9. **Spam rules are rule-based.** They are auditable and reversible (every row carries `filter_reason`) but neither exhaustive nor perfect; small numbers of false positives exist in `promotional_posts.csv` (e.g. genuine news on bookmaker domains) and some spam remains in the master.
10. **Analysis variables are model outputs (v3.0).** `topic_id`, `category_id` and the sentiment scores (Section 4.9) are produced by a topic model, two small language-model agents and a lexicon, respectively. Topic and category ids are only meaningful within a period bin (nine separate models); comparisons across bins need the category crosswalk. Topic granularity differs between the small tournament bins and the large interim bins (a relative minimum topic size). VADER scores headline-length text with a general lexicon and misreads sarcasm and football vocabulary; a validation sample is part of the pipeline. Posts with fewer than three word tokens have no topic or category.

---

8. Methodological use of the data

Full design, settings with rationale, prompts and flagged limitations: `docs/session-04_2026-10-22/01_analytical-pipeline-design.md` (session 04, 2026-10-22; decisions by the author recorded there). Summary:

8.1 Corpus construction for topic modelling. Unit of analysis: one post = one document. Text: the cleaned `title`, a newline, and the cleaned `selftext` truncated to 1,000 characters (cleaning removes Markdown link syntax, URLs, Reddit sprite codes, HTML entities and extra whitespace; it never changes words). Posts with fewer than three word tokens are not modelled. All retained posts are used, including human match/daily threads (`post_type`) and posts whose `lang` is not `en`. The corpus is divided into nine period bins (Section 4.9, `period_bin`) and a separate BERTopic model is fitted per bin: `all-mpnet-base-v2` sentence embeddings → UMAP (15 neighbours, 5 dimensions, cosine, seed 42) → HDBSCAN (minimum topic size 0.25 % of the bin, floor 25, adjusted by a guard to 50–100 topics; `min_samples` 10) → c-TF-IDF keywords (1–2-grams, English and scaffolding stop-words, KeyBERT-inspired re-ranking with maximal marginal relevance). Outliers are reassigned to their nearest topic only above a calibrated embedding-similarity threshold (`topic_assignment`). Each topic is then labelled by two Qwen3-1.7B analyst agents that must agree (multi-turn debate, explicit acceptance, at most five rounds; unresolved topics are flagged for the co-authors), and the labelled topics of each bin are grouped by the same agents into a discourse taxonomy (`category_id`); co-author overrides are recorded in editable tables.

8.2 Sentiment analysis. VADER (Hutto & Gilbert 2014) compound scores: `post_sentiment` on the title (comparable across bins and post types), `post_sentiment_full` on title + body, `post_sentiment_label` at the conventional ±0.05 thresholds. A 300-post sample stratified by bin, label and `is_self` is hand-coded by two co-authors to report agreement. Title-only posts (88 %) are scored on their title; the lexicon's known weaknesses on sports vocabulary and sarcasm are documented in the design (F10).

8.3 Discourse analysis. The coding frame is the per-bin taxonomy (`results/analysis/categories_all_bins.csv`) with its topic labels (`topic_labels_all_bins.csv`); cross-bin comparison of World Cup periods uses a manual crosswalk of categories to chapter-level themes. `post_type`, `lang` and flair are covariates for checking topics (share of match threads, share of English titles per topic), not inputs to the model. Time windows are the period bins; topic prevalence is reported weekly in tournament bins and monthly in interim bins.

8.4 Reproducibility. Scripts `code/06-analysis-pipeline.py` (+ `code/topicmodel/`), `code/06b-analysis-pipeline-colab.ipynb`, `code/07-tag-corpus.R`, `code/08-validate-analysis.R` (session 05). Pinned package versions (`code/requirements-topicmodel.txt`) and model revisions; seed 42; greedy decoding; every run folder holds `config.json` (settings, prompts, backend, versions), the intermediate arrays (UMAP output, raw HDBSCAN labels) and the full agent transcripts, so every downstream output can be replayed exactly.

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9. Version history

Version	Date	Change
3.0 (draft)	2026-10-23	Analysis variables specified (Section 4.9: `period_bin`, `period_type`, `topic_id`, `topic_assignment`, `category_id`, `post_sentiment`, `post_sentiment_full`, `post_sentiment_label`), methodological use documented (Section 8) and limitation 10 added, from the session-04 pipeline design. **The data files remain v2.1 until stage S6 appends the 8 columns to the master and yearly files**; this row is then finalised with the run date and taxonomy version. The pipeline code was implemented and dry-run on bin 1 in session 05 (2026-10-23, `docs/session-05_2026-10-23/`).
1.0	2026-08-09	Raw export (`soccer_posts.csv`, 2,047,322 posts, 64 columns) documented. First cleaned files produced with `data.table::fread()`: 25 columns dropped, posts with `selftext` `[removed]`/`[deleted]` split off (1,591,577 retained).

Version	Date	Change
2.1	2026-10-13	Gambling rule confirmed by the team. Stage S6 now **repairs** encoding damage by strict byte-level re-decoding before filtering (126 posts repaired, 335 filtered); repairs logged in `encoding_repairs.csv` and verified by two new validation checks (28 in total). Clean master: 1,302,021 posts.
2.0	2026-10-11	**Full rebuild from the raw export.** (1) Fixed a quote-doubling corruption introduced by `fread()` in v1 (every literal `"` had become `""`; JSON columns unparseable). (2) Removal detection extended to link posts via `removed_by_category` and placeholder titles. (3) New side files for bot threads, promotional content, encoding issues and non-English titles. (4) Derived columns `post_type` and `lang` added. (5) One writer and one format for all files; redundant duplicate master in the yearly folder removed. (6) Validation script added (26 checks). Details: `docs/session-01_2026-10-11/`.

10. References

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Citation. When using this corpus, cite the Arctic Shift project as the data source and this codebook (version 3.0, McGill University, 2026; version 2.1 for the files without analysis variables).